

HirePulse ATS Platform — Comprehensive Technical Guide

Folder-by-Folder Code Architecture & End-to-End Workflow Manual

1. Executive Overview & System Architecture

HirePulse ATS is an enterprise-grade, multi-tenant Applicant Tracking System & Recruitment SaaS platform. Built with **FastAPI (Python)** on the backend, **MySQL (SQLAlchemy ORM)** for relational database management, and **React (Vite + Tailwind CSS)** on the frontend, the platform provides seamless end-to-end recruitment automation. Key capabilities include multi-tenant organization data isolation, 4-tier Role-Based Access Control (RBAC), automated interviewer slot overlap validation, Google Calendar event synchronization, PDF offer letter generation, and Stripe checkout payments.

Core Platform User Roles & Permissions Matrix

Role Name	Key Responsibilities & Permissions	Accessible Views
Company Admin	Full tenant ownership, manage subscription plans, customize plan prices in INR, monitor organization analytics.	/admin, /subscription
HR Manager	Oversee recruitment metrics, review candidates, inspect interviewer schedules, generate PDF offer letters.	/hr, /recruiter, /subscription
Recruiter	Post job openings, promote jobs (Stripe █1,499), schedule interviews with overlap checks, issue offer letters.	/recruiter
Candidate	Browse job listings, submit job applications, sync interviews to Google Calendar, download & accept offer letters.	/candidate, /job-board

2. Backend Folder & Codebase Breakdown (`backend/`)

The backend is structured into modular layers following clean architecture principles:

Directory Path	Key Files	Technical Purpose & Functionality
<code>backend/app/core/</code>	config.py database.py security.py	Core setup: Reads environment configuration (.env), initializes SQLAlchemy MySQL database engine & sessionmakers, provides bcrypt password hashing and JWT access token creation/verification.
<code>backend/app/models/</code>	models.py	Database ORM Definitions: Defines User, Organization, Job, JobApplication, Interview, OfferLetter, and SubscriptionPlan relational schemas with foreign key constraints.
<code>backend/app/schemas/</code>	schemas.py	Data Validation: Defines Pydantic models for request/response serialization (UserRegister, UserLogin, JobCreate, InterviewScheduleCreate, InterviewUpdate, OfferLetterCreate).

backend/app/services/	stripe_service.py gcal_service.py email_service.py pdf_service.py	External Integrations: • stripe_service: Stripe Checkout for Pro subscriptions & █1,499 job promotions. • gcal_service: Google Calendar 1-click Add URLs & Meet links. • email_service: Invitation & alert emails. • pdf_service: ReportLab PDF generator for Offer Letters & Schedule reports.
backend/app/api/v1/	auth.py jobs.py applications.py interviews.py offer_letters.py subscriptions.py analytics.py	API Controllers: • auth.py: Login, register, profile update, password reset. • jobs.py: Job posting CRUD, public search, Stripe promotion checkout. • interviews.py: Slot overlap prevention, Google Calendar sync, rescheduling. • offer_letters.py: PDF generation, candidate accept/decline. • subscriptions.py: Admin plan price updates & Stripe webhooks.

3. Frontend Folder & Codebase Breakdown (`frontend/src/`)

The frontend is built with React 18, Vite, and Tailwind CSS. Key directory components:

Directory Path	Key Components / Pages	UI Functionality & Design Description
<code>frontend/src/pages/</code>	LandingPage.jsx LoginPage.jsx RegisterPage.jsx AdminDashboard.jsx HRDashboard.jsx RecruiterDashboard.jsx CandidateDashboard.jsx SubscriptionPage.jsx ResetPasswordPage.jsx	Role-Based Views: • LandingPage: Public job board with Featured Premium job cards. • LoginPage / RegisterPage: Authentication with Forgot Password recovery. • AdminDashboard: Multi-tenant organization analytics & dynamic plan price editor. • RecruiterDashboard: Pipeline management, job posting, █1,499 promotion, interview scheduling. • CandidateDashboard: Applications tracker, Google Calendar sync, PDF offer letter response.
<code>frontend/src/components/</code>	Navbar.jsx InterviewCalendarModal.jsx NotificationsModal.jsx ProfileModal.jsx ProtectedRoute.jsx Modal.jsx StatusBadge.jsx	Reusable Modular UI: • Navbar: Top navigation header with Notification Bell badge & enlarged company logo. • InterviewCalendarModal: Calendar view with Google Cal sync & Recruiter reschedule edit form. • NotificationsModal: Dedicated alerts tab for Google Calendar events. • ProtectedRoute: Enforces RBAC route protection.
<code>frontend/src/services/</code> <code>& context/</code>	api.js AuthContext.jsx	State & Data Layer: • api.js: Axios HTTP client configured with JWT Authorization headers. • AuthContext.jsx: Global authentication state, login/logout context provider.

4. Complete End-to-End Workflow Execution

A. Candidate Job Application & Pipeline Movement:

1. Candidate registers account or logs in at `/login`.
2. Candidate searches public jobs on `/job-board` and submits application.
3. Application is created in database with initial status **Applied**.
4. Recruiter reviews application pipeline on `/recruiter` and advances status to **Shortlisted**.

B. Interview Scheduling, Overlap Validation & Google Calendar Sync:

1. Recruiter selects date/time (e.g. 11:00 AM) and submits interview request.
2. Backend checks if interviewer has another interview within ± 29 minutes window. If conflict exists, 400 error is thrown.
3. If valid, backend generates unique Google Meet link and direct 1-click Google Calendar Add URL (`gcal_url`).
4. Automated email invitation with Google Calendar sync button is dispatched to candidate.
5. Recruiter can edit / reschedule interview date/time anytime. Rescheduling updates Google Calendar links and notifies candidate.

C. Offer Letter Generation & Acceptance:

1. Recruiter/HR specifies annual compensation and joining date.
2. ReportLab PDF engine compiles an official Employment Offer Letter PDF stored in `backend/uploads/pdfs/`.
3. Candidate views offer card on `/candidate`, downloads PDF, and clicks **Accept Offer**.

D. Stripe Payment Integrations:

1. **Company Pro Subscription:** Monthly recurring subscription (admin customizable price, e.g. █7,999/mo).

2. **Premium Job Promotion:** One-time ■1,499 Stripe Checkout payment promoting job listing to Featured Premium status.